

رہی تعلیم مشرق میں، نہ مغرب میں رہی غیرت
یہ جتنے پڑھتے جاتے ہیں، جہالت بڑھتی جاتی ہے!

PHY301-Final Term
Subjective Questions from
Past Papers Solved
with References
By Masoom Fairy

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Final Term Examination

Fall 2012

Phy301-Circuit Theory

1. State kirchhoff's current law (KCL)

Marks: 2:

Answer: (PAGE 42)

KIRCHHOF'S CURRENT LAW

Sum of all the currents entering in the node is equal to sum of currents leaving the node. It can also be defined as

$$\text{sum of entering currents} + \text{sum of leaving currents} = 0$$

2. If we want to find V_{out} by superposition, what type of replacement we will do with voltage and current source? Just draw the diagrams.

Answer: (PAGE 92)

“ In any linear circuit containing multiple sources, the current or voltage at any point in the circuit may be calculated as the algebraic sum of the individual contributions of each source acting alone.”

When determining the contributions due to independent sources, any remaining current sources are made zero by replacing them by open circuit and any voltage sources are made zero by replacing them by short circuit.

3. State the superposition theorem.

Answer: (PAGE 92)

The principle of superposition, which provides us with the ability to reduce a complicated problem

to several easier problems – each containing only a single independent source – states that

“ In any linear circuit containing multiple sources, the current or voltage at any point in the circuit may be calculated as the algebraic sum of the individual contributions of each source acting alone.”

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When determining the contributions due to independent sources, any remaining current sources are made zero by replacing them by open circuit and any voltage sources are made zero by replacing them by short circuit.

4. Forward biased or reversed biased diode.

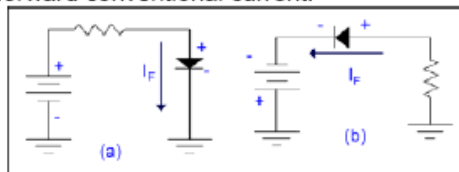
Answer: (PAGE 124 & 125)

THE FORWARD BIASED PN-JUNCTION (DIODE)

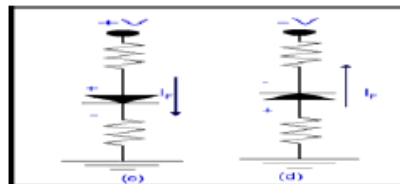
When positive terminal of a battery is connected to the anode or P-side and negative terminal to the cathode or to the N-side of a diode (PN-junction), the diode is said to be **FORWARD-BIASED**.

In the figures to follow it may be noted that the arrow points to the more negative potential in each case. In **fig. a** below the positive terminal of a battery through a resistor is connected to the anode whereas the cathode is attached to the neutral terminal and now the condition of the forward biased is fulfilled.

In **fig. b** below the negative terminal of a battery is connected to the cathode whereas the anode is attached through a resistor to the neutral terminal and now the condition of the forward biased is fulfilled where I_F shows the forward conventional current.



In **fig. c** below the positive volts has been applied to the anode of the diode through a resistance and cathode terminal of the diode is connected to the neutral terminal through resistance which will fulfill the condition of forward biased. Where I_F shows the forward conventional current.



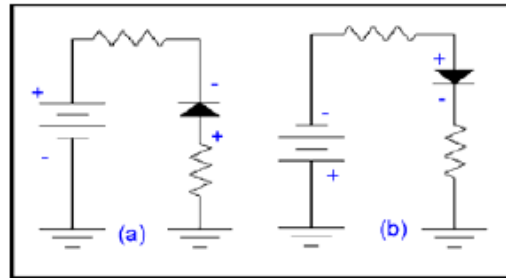
In **fig. d** above the -ve volts has been applied to the cathode of the diode through a resistance and anode terminal of the diode is connected to the neutral terminal through resistance which will fulfill the condition of forward biased. Where I_F shows the forward conventional current will be establish in the diode.

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THE REVERSE - BIASED PN-JUNCTION (DIODE)

When positive terminal of a battery is connected to the cathode or N-side and negative terminal to the anode or to the P-side of a diode (PN-junction), the diode is said to be **REVERSED - BIASED**.

A PN-junction diode is reverse biased when the n type material (cathode) is more positive than the p type material (anode). This causes the depletion region to widen and prevent current. A diode will not conduct when the arrow points to the more positive of the diode potentials.

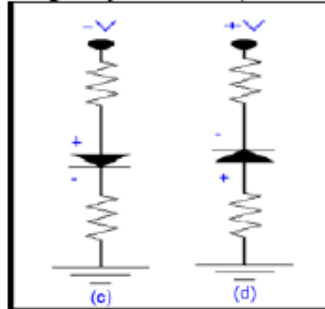


In **fig. a** the negative of the battery is connected to the anode through a resistor and positive terminal of the battery is connected the cathode therefore no current will flow and we can say that diode is not existing and it will act as an open circuit.

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In **fig. b** the negative terminal of the battery is connected to the anode through a resistor and positive terminal of the battery is connected the cathode.

The reverse – biased diode behaves opposite to the forward-biased diode. This means in case of a reverse-biased diode , majority carrier current does not flow and instead of that only minority carrier current can flow. Moreover, the depletion charge layer will expand or enlarged compared to the forward-biased diode where the depletion charge layer shrinks, so it will act as an open circuit.



In **fig. c** the negative voltage has been applied to the anode of the diode through a resistance and cathode terminal of the diode is connected to the neutral terminal through resistance. Which will fulfill the condition of reversed biased.

In **fig. d** the positive voltage has been applied to the cathode of the diode through a resistance and anode terminal of the diode is connected to the neutral terminal through resistance. Which will fulfill the condition of reversed biased

Here we can say that the reverse-biased PN-junction can't support majority carrier current but it will allow the minority carrier current to flow across the junction. This minority carrier current is called as reverse current and is much smaller than the forward majority carrier current of the forward biased PN junction.

5. Differentiate between half wave and full wave rectifier.

Answer:

The difference between a half wave rectifier and full wave rectifier is that a half wave rectifier removes one of the positive or the negative half cycle of the wave and only either half of the cycle appears in the output whereas in the full wave rectifier both the cycles appear in the positive or negative cycle of the output.

The efficiency of a full wave rectifier (81.2%) is too double of a half wave rectifier(40.6%) because the r.m.s. value in case of a full wave rectifier is Maximum current divided by 1.41

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(under root of 2) whereas in case of a half wave rectifier the r.m.s current is half of maximum current during the wave cycle.

<http://www.indiastudychannel.com/experts/18455-what-Differences-between-half-wave-full-wave.aspx>

OR

Half Wave Rectifier: A half wave rectifier is a special case of a clipper. In half wave rectification, either the positive or negative half of the AC wave is passed easily, while the other half is blocked, depending on the polarity of the rectifier. Because only one half of the input waveform reaches the output, it is very inefficient if used for power transfer. Half-wave rectification can be achieved with a single diode in a one phase supply.

Full Wave Rectifier: A full-wave rectifier converts the whole of the input waveform to one of constant polarity (positive or negative) at its output. Full-wave rectification converts both polarities of the input waveform to DC (direct current), and is more efficient. However, in a circuit with a non-center tapped transformer, four diodes are required instead of the one needed for half-wave rectification.

(From other past papers)

6. Conduction band and valence band.

Answer:

Valence band: <http://www.answers.com/topic/valence-band>

Valence band is the highest range of electron energies in which the electrons are normally present at absolute zero temperature.

Conduction band: http://en.wikipedia.org/wiki/Conduction_band

This is the range of electron energies enough to free an electron from binding with its atom to move freely within the atomic lattice of the material.

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7. Why is electron flow with the arrow in the symbol of a Zener diode instead of against the arrow as it is in a normal diode?

Answer:

Because Zener diodes are operated in the reverse bias mode.

<http://quizlet.com/15560197/navedtra-14179-chapter-3-flash-cards/>

8. What resistance property is found in tunnel diodes but not in normal diodes?

Answer:

Negative resistance is found in tunnel diodes which made them different.

<http://www.tpub.com/neets/book7/26m.htm>

9. Differentiate between Clippers & Clampers. Also write the various types of Clipper circuits and define each type.

Answer:

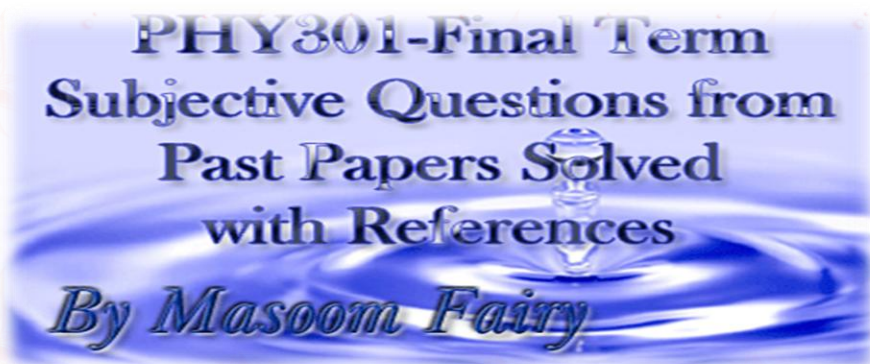
(PAGE 160 and 161)

Clippers,

There are variety of diode networks call clippers that have the ability to clip off a portion of the input signal without distortiong the remaining part of the alternating waveform. The half wave rectifier studied earlier is a simplest form of diode clipper. Depending on the orientation of the diode the positive or negative region of the input signal is clipped off.

It makes a sketch in mind about the response of the network.determine the applied voltage that causs change in the diode bise. be continuously aware of the difined terminal and porlarity of V_o . Sketch the input signal on the top and the output at the bottom to determine the output at instantaneouse pointer of the input.

Types of Clipper,



There are two general types

Series

The series is defined as one where the diode is in series with the load as half wave rectifier.

Parallel

The parallel is the one where the diode is in parallel with the load as half wave rectifier.

Clampers,

There are the circuits which clamp the input signal to a different level depending upon the configuration of the clamper circuit.

10. What is the dependence of junction area and temperature of pn junction diode on Leakage current?

Answer: (PAGE 132)

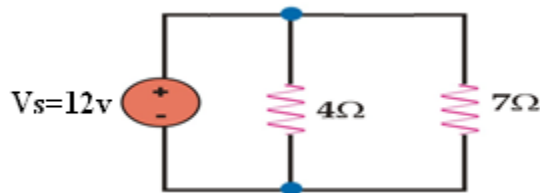
The leakage currents are proportional to the junction area, just as I_s . Its dependence on temperature is however, different from that of I_s .

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1. Find the voltage drop at 4 ohms. [marks 2]



Answer: (Solved according to PAGE 32)

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The Formula is $V = R_1 * V_s / R_1 + R_2$

By putting values,

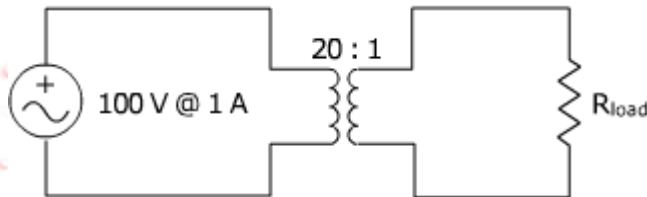
$$= 4 * 12 / 4 + 7$$

$$= 48 / 11$$

$$= 4.36 = \text{Ans.}$$

2. What would be the current across the load?

[marks3]



Answer:

Formula is $I_2 = N_2 / N_1 * I_1$

$$= 1 / 20 * 1 \text{ A}$$

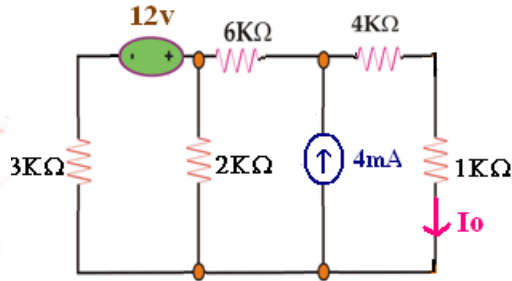
$$= 0.05 \text{ A}$$

$$= 50 \text{ mA} = \text{Ans.}$$

3. Keeping in mind Norton's Theorem find Norton's Resistance R_N . Do not draw the circuit. Write each step of calculation.

[Marks5]

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Answer:

First Step:

Remove the load resistance R_L .

Second Step:

Find I_{in} by short circuiting the open terminals of the circuit.

$$I_{in} = I_3 = 4\text{mA}$$

Third Step:

Find R_n by short circuiting all voltage sources.

$$3\text{k}\Omega \parallel 2\text{k}\Omega = \frac{3 \times 2}{3+2} = 1.2\text{k}$$

$$1.2\text{k} + 6\text{k} + 4\text{k} = 11.2 = \text{Ans.}$$

4. What is rectification and amplification, and what devices use rectification and amplification? [marks5]

Answer:

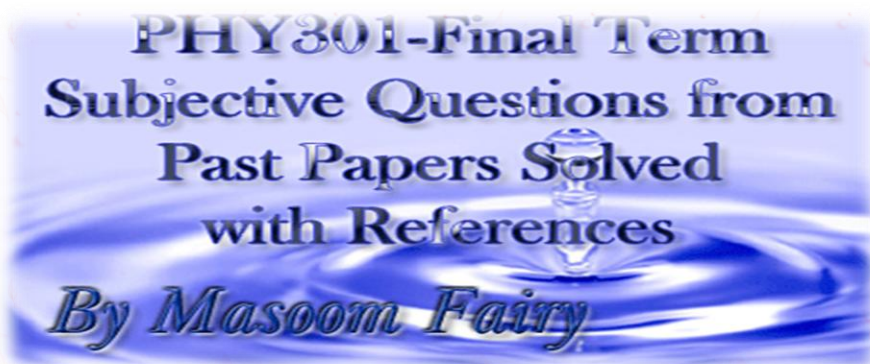
Rectification is the process of converting an AC signal into a DC signal. It carried out at all levels of electric power, from a thousandth of a watt to detect an AM radio signal, to thousands of kilowatts to operate heavy electric machinery.

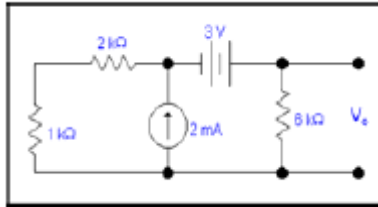
Devices are Mobile phones, VCRs, washing machines etc.

Amplification is the process of increasing the power of the signal.

Devices are radio receivers, oscillators etc.

5. Calculate the voltage V_o by using Thevenin's theorem .

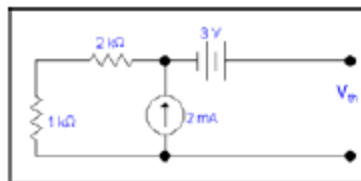




Answer: (PAGE 100 and 101)

We want to calculate V_o by using Thevenin's theorem .we will follow the steps given earlier.

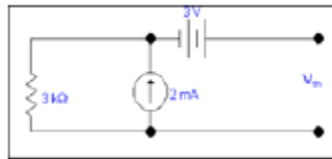
First step: Removing R_L



Here R_L is 6k resistor at which we want to calculate the voltage V_o .

Second step: Calculating V_{th}

1k is in series with 2k so modified circuit will be as



3k is in parallel with 2mA source, so by source transformation

Now 2mA source is in parallel with 3k resistor. So it can be changed to a voltage source of value = $2m \times 3k$ (by ohm's Law)
= 6 Volts.

3k resistor will become in series with this source as shown in the circuit below

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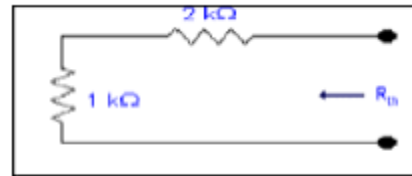
Now the combined effect of these two source will be 9 volts.



So

$$V_{th} = 9 \text{ volts}$$

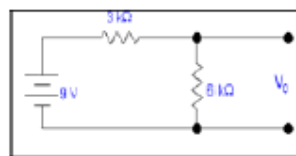
Third step: Calculating R_{th}



$$R_{th} = 3k$$

Fourth step: Calculating the unknown quantity.

After calculating V_{th} and R_{th} , re-inserting the load resistance R_L in the circuit in series with R_{th} and considering the V_{th} as a battery in series with these two resistances.



$$V_0 = (6k/9k) \times 9$$

$$V_0 = 6 \text{ volts}$$

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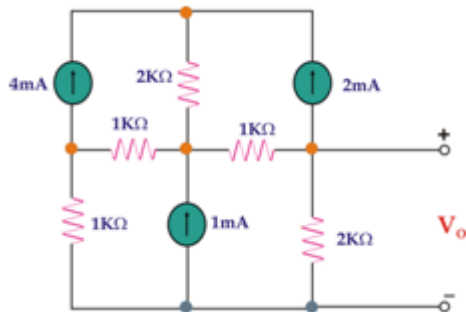
1. What is Zener diode? [2 marks]

Answer:

A Zener diode is a type of diode that permits current not only in the forward direction like a normal diode, but also permits in the reverse direction when the voltage is above a certain value known as the breakdown voltage, “zener knee voltage” or “zener voltage” or “avalanche point”.

http://en.wikipedia.org/wiki/Zener_diode

2. Using the **Thevenin's theorem**, what type of changes we will do in the circuit, to find Thevenin's voltage V_{th} ? No need to solve the circuit. [2 marks]

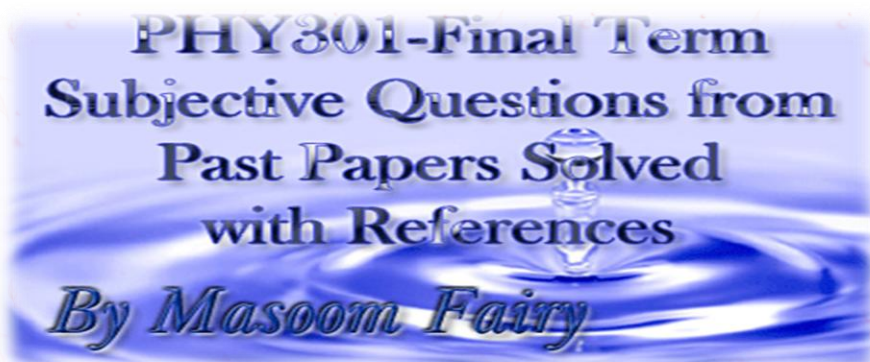


Answer: (LECTURE 25)

We perform only two steps for finding V_{th} . In first step we have to remove Load resistance which is 2k ohms then in second step we have to Calculate V_{th} by using any method at open terminals of open circuit.

3. State Kirchhoff's voltage law (KVL). [2 marks]

Answer: (PAGE 58)



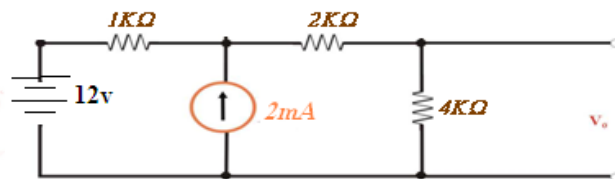
KIRCHHOFF'S VOLTAGE LAW:

This law states that the algebraic sum of the voltages around any loop is zero.

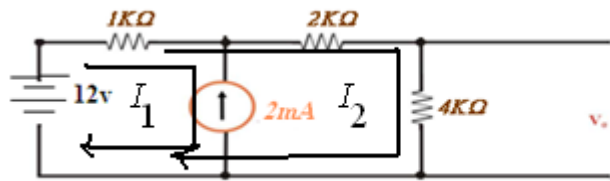
OR

Sum of voltages rises and voltage drops around any closed path or loop is equal to zero.

4. Calculate V_0 with the using KVL equation. [5 marks]



Answer:



$$I_2 = -2mA$$

$$4KI_2 + 2KI_2 + 1K(I_2 + I_1) = 12$$

$$7KI_2 + 1KI_1 = 12$$

$$7KI_2 - 2 = 12$$

$$I_2 = 14/7$$

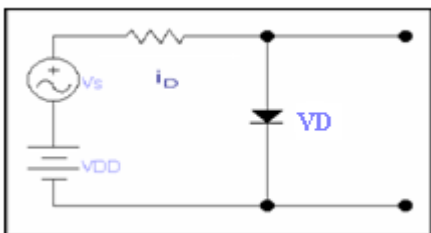
$$I_2 = 2mA$$

$$V_0 = 4k(2mA)$$

$$= 8volts = Ans$$

5. Consider the circuit as shown, Write V_s equ. for AC analysis purposes. [3 marks]

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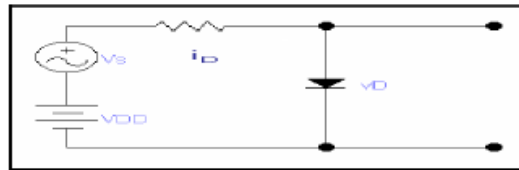


Answer:

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APPLICATION:

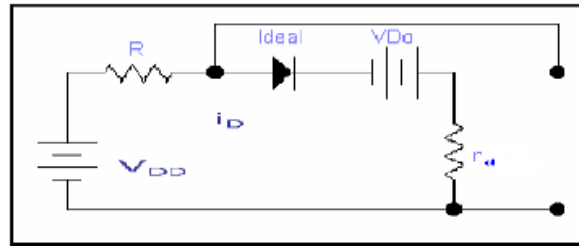
Consider the circuit as shown, for analysis purposes, we can split the circuit into two parts that is ac and dc



DC source having the value of V_{DD}

we replace the ac and replace diode with constant drop model

For dc analysis the Circuit will be



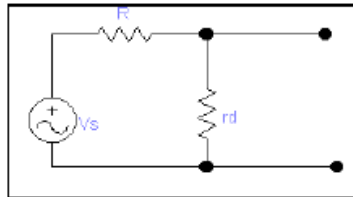
in case of DC we consider only DC current I_D is flowing and no effect of the ideal diode because this ideal diode is forward biasing and it results in short circuit therefore

From KVL

$$V_{DD} - I_D R - V_{Do} - I_D r_d = 0$$
$$V_{DD} = I_D R + V_{Do} + I_D r_d \text{ ----- (A)}$$

in case of DC only DC current I_D is flowing the

For ac analysis we will remove DC sources which are the part of the original circuit and also remove the DC source which appear in the previous effective circuit and we also remove the ideal circuit so our circuit will be as



For ac analysis

$$v_s = i_d (R + r_d) \text{ ----- (B)}$$

by combining (A) and (B)

Overall analysis is

$$V_{DD} + v_s = I_D R + V_{Do} + I_D r_d + i_d (R + r_d)$$
$$= I_D (R + r_d) + i_d (R + r_d) + V_{Do}$$
$$V_{DD} + v_s = (R + r_d)(I_D + i_d) + V_{Do} \text{ ----- (C)}$$

But



$$I_D + i_d = i_D$$

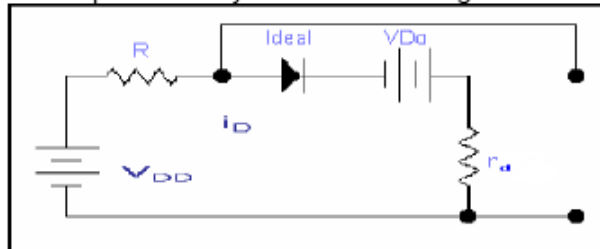
therefore

$$V_{DD} + v_s = (R + r_d)(I_D + i_d) + V_{D0}$$

separating the dc and signal quantities on both sides of equation (C)

$$V_{DD} = I_D R + V_{D0}$$

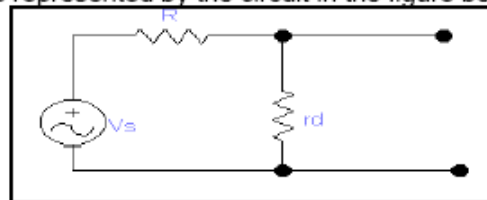
which is represented by the circuit in the figure below



and for the signal

$$v_s = i_d (R + r_d)$$

which is represented by the circuit in the figure below



However, if we carefully see the ac equation circuit, it is nothing more than a voltage divider. Hence the diode signal voltage will be

$$V_d = V_s r_d / (r_d + R)$$

6. Describe the function of transistor. [3 marks]

Answer:

In electronics, a transistor is a semiconductor device commonly used to amplify or switch electronic signals. A transistor is made of a solid piece of a semiconductor material, with at least three terminals for connection to an external circuit. A voltage or current applied to one pair of the transistor's terminals changes the current flowing through another pair of terminals. Because the controlled power can be much larger than the controlling power, the transistor provides amplification of a signal.

<http://answers.yahoo.com/question/index?qid=20080916043112AAra3iv>

7. What is the difference between Ideal diode model and practical diode model?

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[3 marks]

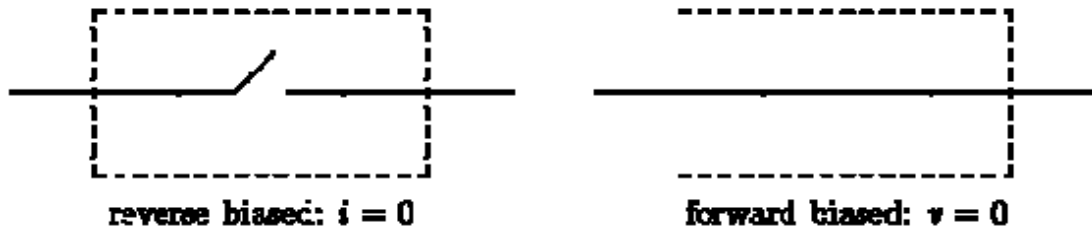
Answer:

(PAGE 136)

Ideal Model:

We have seen that the diode behaves essentially as a **switch**: on when forward biased, off when reverse biased.

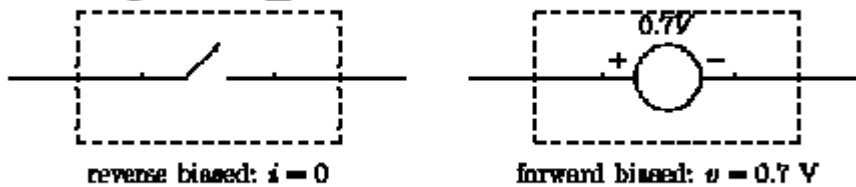
Figure Ba: Ideal diode model.



PRACTICAL MODEL:

In practice we find that there is a voltage drop of about 0.7 V across the diode (silicon; germanium is 0.3V) when it is forward biased, and so it is often useful to include this voltage drop in circuit analysis.

Figure C_a: Practical diode model.



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Question No: 1 (Marks: 2)

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In the NPN transistor, what section is made very thin compared with the other two sections? (Marks: 2)

Answer: (PAGE 175)

The base layer is very thin as compared to other two, normally in the range of 150:1.

Question No: 32 (Marks: 2)

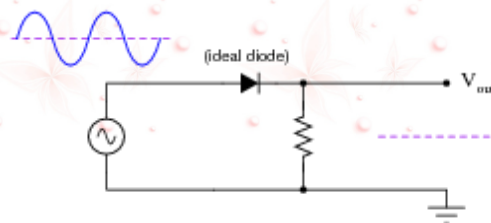
In a reverse biased PN-junction, which current carriers cause leakage current?

Answer: (Video Lecture 30)

Doping current carriers cause leakage current.

Question No: 33 (Marks: 2)

Sketch the shape of the output voltage waveform for this "clipper" circuit, assuming an ideal diode with no forward voltage drop:



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Answer: (Not conform)

It will be step sin wave or half sin wave or half wave

Question No: 34 (Marks: 3)

Find the value of I_E for transistor if $I_B = 1.5\text{mA}$ and $I_C = 4\text{mA}$

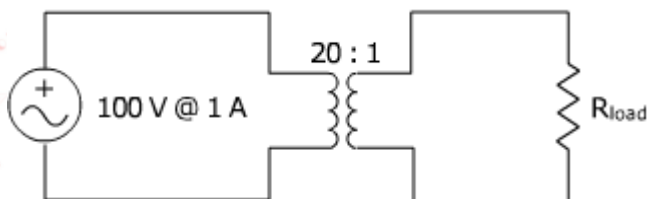
Answer: (Lecture 44)

$$I_e = I_b + I_c$$

$$\text{So, } 1.5 + 4 = 5.5 = \text{Ans.}$$

Question No: 35 (Marks: 3)

What would be the current across the load?



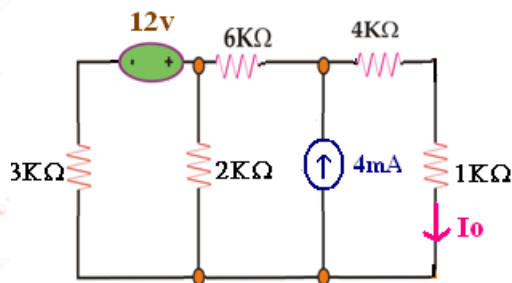
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Answer: (REPEATED)

$$\begin{aligned}\text{Formula is } I_2 &= N_2/N_1 \cdot I_1 \\ &= 1/20 \cdot 1\text{A} \\ &= 0.05\text{A} \\ &= 50\text{mA} = \text{Ans.}\end{aligned}$$

Question No: 36 (Marks: 3)

Keeping in mind **Norton's Theorem** find Norton's Resistance R_N . Do not draw the circuit. Write each step of calculation.



Answer: (REPEATED)

First Step:

Remove the load resistance R_L .

Second Step:

Find I_{in} by short circuiting the open terminals of the circuit.

$$I_{in} = I_3 = 4\text{mA}$$

Third Step:

Find R_n by short circuiting all voltage sources.

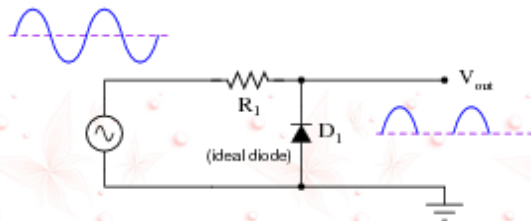
$$3\text{k} \parallel 2\text{k} = 3 \cdot 2 / (3 + 2) = 1.2\text{k}$$

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$$1.2k+6k+4k = 11.2 = \text{Ans.}$$

Question No: 37 (Marks: 5)

Predict how the operation of this clipper circuit will be affected as a result of the following faults. Consider each fault independently (i.e. one at a time, no multiple faults):



- a) Diode D_1 fails open:
- b) Diode D_1 fails shorted:
- c) Resistor R_1 fails open:
- d) Resistor R_1 fails shorted:

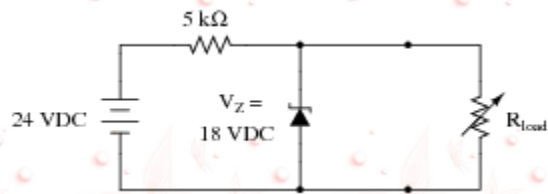
For each of these conditions, explain *why* the resulting effects will occur.

Answer:

Question No: 38 (Marks: 5)

At what load resistance value will this voltage regulator circuit begin to lose its ability to regulate voltage? Also, determine whether the voltage regulation is lost for load resistance values greater than this threshold value, or less than this threshold value.

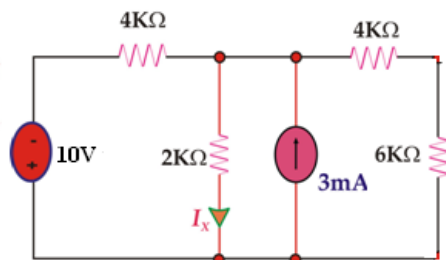
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Answer:

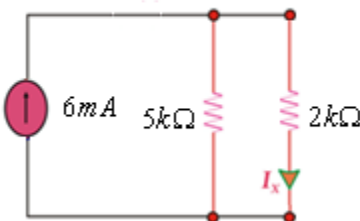
Question No: 39 (Marks: 5)

Find I_x by **Norton's Theorem** if **Norton's Resistance R_N** is $5k\Omega$ and **Norton's current I_{Nor}** is $6mA$, draw the **Norton's equivalent** circuit.



Answer:

Norton's equivalent circuit,



$$I_x = R_2 * I_t / R_1 + R_2$$

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$$= 5k \cdot 6 / 2k + 5k = 30/7 = 4.28 = \text{Ans.}$$

Final Term Examination

Fall 2009

Phy301-Circuit Theory

Question No: 1 (Marks: 3)

A small light bulb with a resistance of 25Ω is connected across the same 220 volt power line. How much is the current 'I'.

Answer:

$R=25\Omega$, $V=220\text{ V}$

From ohm law $V=IR$,

$I=V/R$ So,

$I=220/25=8.8\text{ Ampere} = \text{Ans.}$

Question No: 2 (Marks: 3)

Describe Source Transformation method for simplifying circuit.

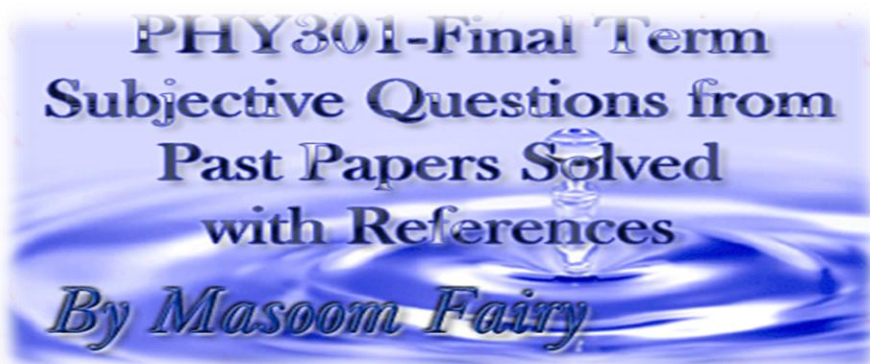
Answer:

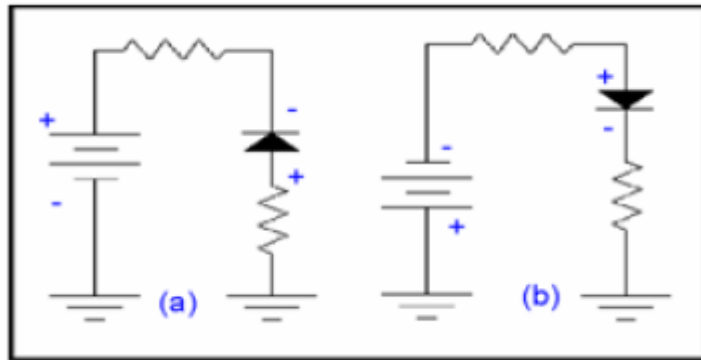
(PAGE 95)

Source Transformation: If we have any source embedded within a network, say this source is a current source having a value I & there exists a resistance having a value R , in parallel to it. We can replace it with a voltage source of value $V=IR$ in series with same resistance R . The reverse is also true that is a voltage source V , in series with a resistance R can be replaced by a current source having a value $I= V/R$ in parallel to the resistance R . Parameters within circuit, for example an output voltage remain unchanged under these transformations.

Question No: 3 (Marks: 3)

Given below are two figures (a) and (b) having Diode, which diode is forward biased or reversed biased, tell reason.



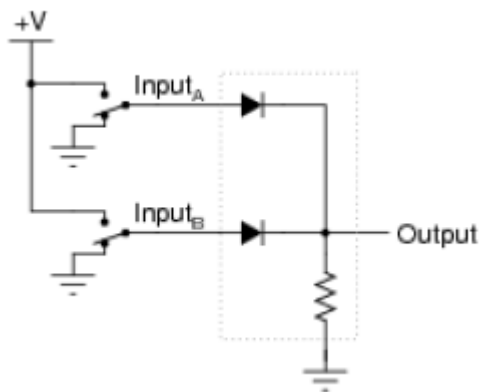


Answer: (PAGE 124)
Both A and B are reversed biased.

In **fig. a**, the negative of the battery is connected to the anode through a resistor and positive terminal of the battery is connected the cathode therefore no current will flow and we can say that diode is not existing and it will act as an open circuit. In **fig. b** the negative terminal of the battery is connected to the anode through a resistor and positive terminal of the battery is connected the cathode.

Question No: 4 (Marks: 3)

Crude logic gates circuits may be constructed out of nothing but diodes and resistors. Take for example this logic gate circuit:



Identify what type of logic function is represented by this gate circuit.

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Answer:

This is an OR gate circuit.

If two diode OR logic gates are cascaded, they behave as current-sourcing logic gates: if the first gate produces high output voltage, the second gate consumes current from the first one. If the first gate produces low output voltage, the second gate does not inject current into the output of the first one. A diode OR gate does not use its own power supply. The input sources with high voltage supply the load through the forward-biased diodes.

http://en.wikipedia.org/wiki/Diode_logic#Diode_logic_gate_versions

Question No: 5 (Marks: 3)

Differentiate between Half wave & Full Wave Rectifier.

Answer: (Repeated)

The difference between a half wave rectifier and a full wave rectifier is that a half wave rectifier removes one of the positive or the negative half cycle of the wave and only either half of the cycle appears in the output whereas in the full wave rectifier both the cycles appear in the positive or negative cycle of the output.

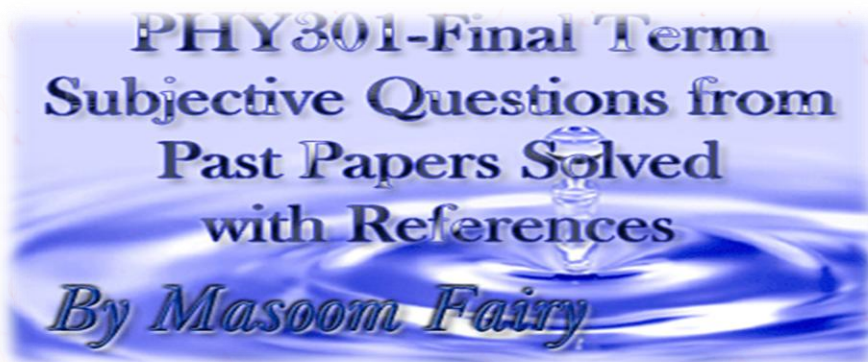
The efficiency of a full wave rectifier (81.2%) is too double of a half wave rectifier (40.6%) because the r.m.s. value in case of a full wave rectifier is Maximum current divided by 1.41 (under root of 2) whereas in case of a half wave rectifier the r.m.s current is half of maximum current during the wave cycle.

<http://www.indiastudychannel.com/experts/18455-what-Differences-between-half-wave-full-wave.aspx>

OR

Half Wave Rectifier: A half wave rectifier is a special case of a clipper. In half wave rectification, either the positive or negative half of the AC wave is passed easily, while the other half is blocked, depending on the polarity of the rectifier. Because only one half of the input waveform reaches the output, it is very inefficient if used for power transfer. Half-wave rectification can be achieved with a single diode in a one phase supply.

Full Wave Rectifier: A full-wave rectifier converts the whole of the input waveform to one of constant polarity (positive or negative) at its output. Full-wave rectification converts both polarities of the input waveform to DC (direct current), and is more efficient. However, in a

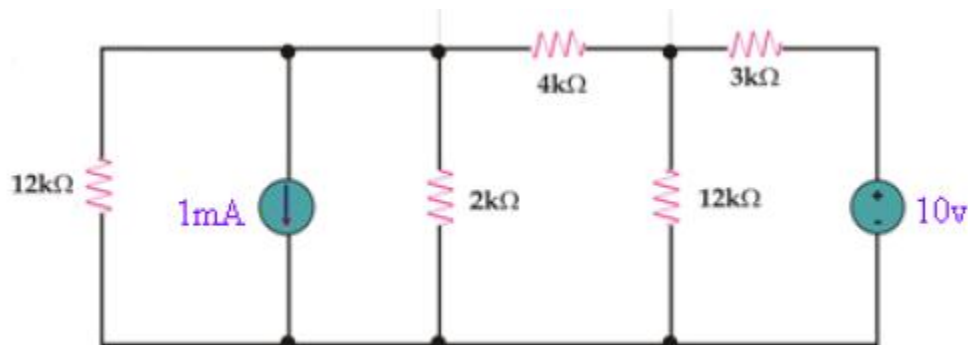


circuit with a non-center tapped transformer, four diodes are required instead of the one needed for half-wave rectification.

(From other past papers)

Question No: 6 (Marks: 5)

Keeping in mind the **Source transformation method**, how we will convert **10v** voltage source into current source and **1mA** into voltage source in the following circuit, Draw diagrams of converted circuit.



Answer:

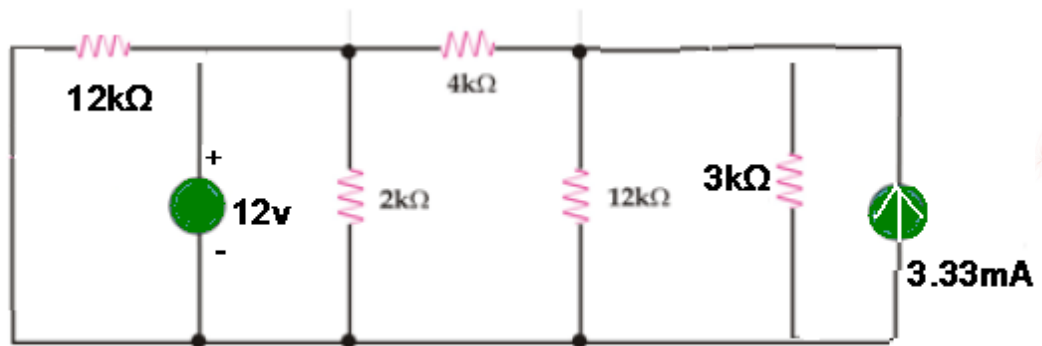
Source transformation method: $I = V/R$

$$= 10/3k = 3.33$$

Then, $V = IR$

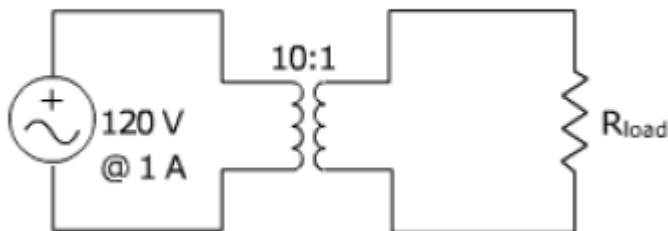
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$$= -1\text{mA} \cdot 12\text{k} = -12\text{v}$$



Question No: 7 (Marks: 5)

Did the power remain same or not, in either case provide a proof?



Answer:

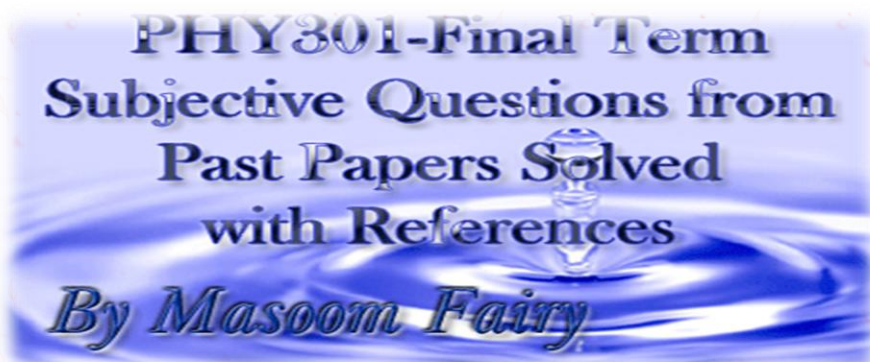
$$P_1 = P_2$$

$$P = VI$$

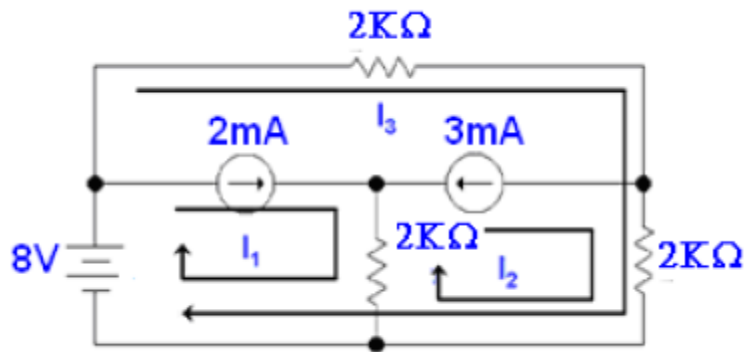
$$V_2 = N_2/N_1 \cdot V_1 = 1/10 \cdot 120 = 12\text{v}$$

SO, the power is not remain same because V_1 is not equal to V_2 .

Question No: 8 (Marks: 5)



Calculate I_1 , I_2 and I_3 values. Using KVL method.



Answer:

$$I_1 = 2\text{mA} \dots \text{Ans}$$

$$I_2 = -3\text{mA} \dots \text{Ans}$$

KVL for loop 3

$$2kI_3 + 2k(I_3 + I_2) = 8$$

$$2kI_3 + 2kI_3 + 2kI_2 = 8$$

$$4kI_3 - 8 = 8$$

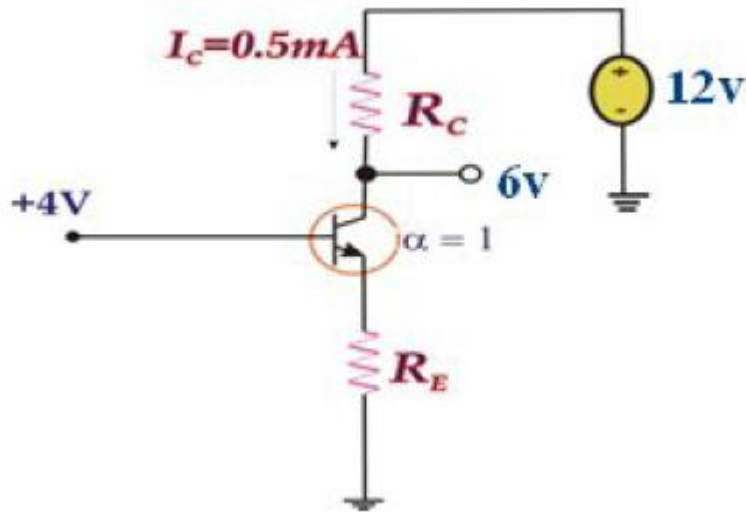
$$4kI_3 = 16$$

$$I_3 = 4\text{mA} \dots \text{Ans}$$

Question No: 9 (Marks: 5)

For the circuit shown, find RE & RC

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Answer: (According to LECTURE 45 (Example 2))

$I_C = 0.5\text{mA}$ and $\alpha = 1$
 $V_{CB} = 2\text{V}$
 $(12-6)/0.5\text{mA}$
 $R_C = 0\text{k}\Omega = \text{Ans.}$

If $\alpha = 1$. Then
 $I_C = I_E = 0.5\text{mA}$
 $4-0.7/0.5$
 $R_E = 2.6\text{k}\Omega = \text{Ans.}$

Question No: 10 (Marks: 5)

A particular diode, for which $n=1$, is found to conduct 3mA with a junction voltage of 0.7V to 0.8V . Find diode current.

Answer: (Page 142)

$I_D = I_{SE} V_D / nV_T$
 If $I_D = 3 \times 10^{-3} \text{A}$
 when $V_D = 0.7$
 then $3 \times 10^{-3} = I_{SE} 0.7 / 25.2 \times 10^{-3}$

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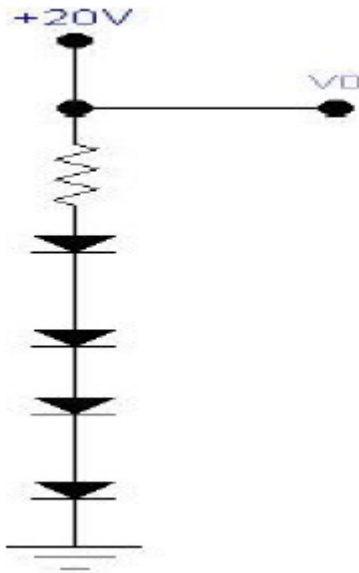
$$= I_s = 2.5905 \times 10^{-15}$$

when $V_d = 0.7$ $I_d = 4.4619 \text{ mA}$

when $V_d = 0.8$ $I_d = 0.15868 \text{ mA}$

Question No: 11 (Marks: 10)

Design the circuit in the fig. so that $V_0 = 3 \text{ V}$ when $I_L = 0$, and V_0 changes by 40 mV per 1 mA of load current. Find the value of R . (assume four diodes are identical) relative to a diode with 0.7 V drop at 1 mA current. Assume $n = 1$



Answer: (PAGE 143)

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$V_0 = 3V$, when $I_L = 0$, therefore each diode should exhibit a drop of $0.75V$. If $I_L = 1mA$, then V_0 changes by $40mV$ and a change due to each diode is $10mV$.

Hence

$$r_d = 10mV/1mA \\ = 10 \text{ Ohms}$$

but

$$r_d = nV_T / I_D \\ 10 = 1 \times 25m / I_D \\ I_D = 2.5mA$$

Hence

$$15 - 3 - I_D R_D = 0 \\ R = (15 - 3) / I_D \\ = (15 - 3) / 2.5m \\ = 4.8k \text{ Ohms.}$$

رہی تعلیم مشرق میں، نہ مغرب میں رہی غیرت
یہ جتنے پڑھتے جاتے ہیں، جہالت بڑھتی جاتی ہے!

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PHY301 - Circuit Theory Glossary By www.virtualians.pk

Ammeter : An instrument designed to read the current through elements in series with the meter.

DC current source : A source that will provide a fixed current level though the load to which it is applied may cause its terminal voltage to change

DC voltage source : A source that will provide a fixed amount of voltages across its terminal regardless of the current passing through the circuit.

Direct current : Current having a single direction (unaltered) and a fixed magnitude over time.

Free Electron : An electron unassociated with any particular atom, relatively free to move through a crystal lattice structure under the influence of external field

Conductance : A measure of the relative ease with which a current can be established in a material.

Short Circuit : A circuit having no load resistance

Open Circuit : A circuit having practically infinit resistance

Circuit Breaker : A two-terminal device designed to ensure that current levels do not exceed safe levels. If 'triped' it can be reset with a switch or a reset button.

Power : Its a measure of energy used or supplied per unit time.

Branch : The current path between any two nodes is called a branch.

Series Resistors : Two current carrying resistors are considered to be in series if they have only one node in common, and that node is not connected to any other current carrying resistor

Circuit : Its a closed path for electric current

Ampere(A) : The SI unit of measurement applied to the flow of charge through a conductor.

Cell : A fundamental source of electrical energy developed through the conversion of chemical or solar energy.

Conductor : Materials that permit a generous flow of electrons with very little voltage applied.

Coulomb(C) : The fundamental SI unit of measure for charge. It is equal to the charge carried by 6.242×10^{18} electrons

Potential Difference : The algebraic difference in potentials between the two points in the electrical system.

Potential Energy : The energy possessed by any mass due to its position in a force field.

Semiconductors : A material having a conductance value between that of conductors and insulators

Ductility : The property of a material that allows it to be drawn into long, thin wires.

Malleability : The property of a material that allows it to be worked into many different shapes.

Resistance : A measure of the opposition to the flow of charge through a material.

Resistivity : It is a characteristic resistance offered by certain material in the path of flowing charges.

Node : A point where two or more circuit elements are connected.

Conventional Current Flow : Conventionally current is considered to flow from a positive potential to the negative potential. It is called conventional current flow.

Internal Resistance : The inherent resistance found internal to any source of energy.

Alpha : Characteristic of junction transistors. Ratio of collector current to emitter current. Value is 0.98 to 0.99

Battery : An energy source that uses a chemical reaction to convert chemical energy into electrical energy. OR Group of cells connected in series or parallel

Capacitor : Device used to store electric charge. Consisting of two conducting plates separated by an insulating material.

Depletion region : The area near a pn junction on both sides that has no majority carriers.

Germanium : A semiconductive material.

Galvanometer : Measures electric charge or current.

Ground : Common return to earth for ac power lines. Chassis ground in electronic equipment is the common return to one side of the internal power supply.

Giga : Metric prefix for

10⁹

LED : Light emitting diode

Load : Takes current from the voltage source, resulting in load current.

Loop : In a circuit, any closed path.

Linear relation : Straight line graph between two variables. As one increases, the other increases in direct proportion.

mks : Meter-Kilogram-Second system of units.

Majority carriers : The most numerous charge carrier in a doped semiconductor material (either free electrons or holes).

Norton's theorem : Method of reducing a complicated network to one current source with shunt resistance.

Neutron : Particle without electric charge in the nucleus of an atom.

Nucleus : The central part of an atom containing protons and neutrons.

Ohm : Unit of resistance. Value of one ohm allows current of one ampere with potential difference of one volt.

Peak value : Maximum amplitude, in either polarity, 1.414 times rms value for sine wave V or I.

PN Junction : The boundary between two different types of semiconductor material.

Power supply : A circuit that converts ac line voltage to dc voltage and supplies constant power to operate a circuit.

Proton : Particle with positive charge in the nucleus of an atom.

Volt : Practical unit of potential difference. One volt produces one ampere of current in a resistance of one ohm.

Voltage source : Supplies potential difference across two terminals.

Beta : Current gain characteristic of junction transistors. Ratio of collector current to base current.

Z : Symbol for ac impedance. Includes resistance with capacitive and inductive reactance.

Y network : Another way of denoting a wye network.

Alternating Current (ac) : Current that reverses direction at a regular rate in response to a change in source voltage polarity.

Atom : The smallest particle of an element possessing the unique characteristics of that element.

Atomic Number : The number of protons in a nucleus.

Amplifier : A device that increases the amplitude of a signal.

Average value : The average of a sine wave over one half cycle. It is 0.637 times the peak value.

Bias : The application of a dc voltage to an electronic device to produce a desired mode of operation.

Bipolar transistor : NPN or PNP type

Charge : An electrical property of matter that exists because of an excess or a deficiency of electrons. Charge can be either positive or negative.

Ceramic : Insulator with a high dielectric constant.

Chassis ground : common return for all electronic circuits mounted on one metal chassis or PC board. Usually connects to one side of dc supply voltage.

Chip : Miniature semiconductor for integrated circuit.

Closed circuit : A circuit with a complete current path.

Delta network : Three components connected in series in a closed loop. Same as pi network.

Current divider : A parallel circuit to divide branch I less than the main -line current.

Current source : Supplies $I = V/r_i$ to load with r_i in parallel.

Dielectric : Insulating material. It cannot conduct current but does store charge.

Diode : Electronic device with two electrodes. Allows current flow in only one direction.

Doping : Adding impurities to pure semiconductor material to provide free positive and negative charge.

Electron : Basic particle of negative charge. In orbital rings around the nucleus in an atom.

Electronics : Based on electrical effects of the electron.

Emf : Electromotive force, voltage to produce current in a circuit.

Emitter follower : Circuit in which signal input is to base and output is from emitter. Same as common -collector circuit.

Farad : Unit of capacitance. Value of one farad stores one coulomb of charge with one volt applied.

FET : Field effect transistor. A device that depends on an electric field to control the current in a silicon channel.

Flux : Magnetic lines of force.

Forward voltage : Polarity that allows current of majority carriers through a semiconductor junction.

Fuse : Metal link that melts from excessive current and opens circuit.

Full- wave rectifier : A circuit that converts an ac sinusoidal input voltage into a pulsating dc voltage with two output pulses occurring for each input cycle.

Henry : Unit of inductance. Current change of one ampere per second induces one volt across an inductance of one henry.

Hertz(Hz) : Unit of frequency. One hertz equals one cycle per cycle.

Hole : The absence of an electron in the valence band of an atom.

Insulator : A material that does not conduct current.

Intrinsic : The pure or natural state of a material.

Inductance(L) : Ability to produce induced voltage when cut by magnetic flux. Unit of inductance is the henry(H).

Ion : Atom or group of atoms with net charge. Can be produced in liquids, gases, and doped semiconductors.

Inductor : Coil of wire with inductance.

Joule(J) : Practical unit of work or energy. One joule equals one watt-second of work.

Kirchhoff,s current law(KCL) : Kirchhoff,s current law(KCL) The phasor sum of all currents into and out of any branch point in a circuit must equal zero.

Kirchhoff,s voltage law(KVL) : The phasor sum of all voltages around any closed path must equal zero.

Mesh current : Assumed current in a closed path, without any current division, for application of kirchhoff,s current law.

Q-point : The dc operating(bias) point of an amplifier specified by voltage and current values.

Zener diode : A diode designed for limiting the voltage across its terminal in reverse bias.

Zener breakdown : The lower voltage breakdown in a zener diode.

Y : Symbol for admittance in an ac circuit. Reciprocal of impedance Z the $Y=1/Z$

XL : Inductive reactance , equal to $2\pi fL$

p

fL

Watt hour : Unit of electric energy, as power x time

Xc : Capacitive reactance, equal to $1/\omega C$

p

fc

Watt(W) : Unit of real power.

Work : Corresponds to energy. Equal to power x time. Basic unit is one joule, equal to one volt - coulomb or one watt second.

Voltage divider : A series circuit to provide v less than the source voltage.

Voltage drop : Voltage across each component in a series circuit. The proportional part of total applied voltage.

Voltage regulator : A device that maintains a constant output voltage with changes in input voltages or output load current.

Balanced Forces : When a number of forces act on a body, and the resultant force is zero, then the forces are said to be resultant forces.

Balmer series : A set of four line spectra, narrow lines of color emitted by hydrogen atom electrons as they drop from excited states to the ground state

Barometer : An instrument that measures atmospheric pressure, used in weather forecasting and in determining elevation above sea level

Beat : Rhythmic increases and decreases of volume from constructive and destructive interference between two sound waves of slightly different frequencies

Bar : A unit of pressure, equal to 105 Pascals.

Beta particle : An electron emitted from a nucleus in radioactive decay.

Bernoulli's theorem: : The total energy per unit volume of a non-viscous, incompressible fluid in a streamline flow remains constant

Binding energy: : The net energy required to decompose a system into its constituent particles

Black body: : An ideal body which would absorb all incident radiation and reflect none

Big bang theory: : Current model of galactic evolution in which the universe was created from an intense and brilliant explosion from a primeval fireball

Black hole: : Black hole: The remaining core of a supernova that is so dense that even light cannot escape.

Blackbody radiation: : Electromagnetic radiation emitted by an ideal material (the blackbody) that perfectly absorbs and perfectly emits radiation

Bohr model: : Model of the structure of the atom that attempted to correct the deficiencies of the solar system model and account for the Balmer series

Boyle's law: : For a given mass of a gas at constant temperature, the volume of the gas is inversely proportional to the pressure.

Brownian motion: : The continuous random motion of solid microscopic particles when suspended in a fluid medium due to the consequence of ongoing bombardment by atoms and molecules.

Bulk's modulus of elasticity: : The ratio of normal stress to the volumetric strain produced in a body.

Buoyant force: : upward force on an object immersed in fluid.

British thermal unit: : The amount of energy or heat needed to increase the temperature of one pound of water one degree Fahrenheit (abbreviated Btu)

Junction: : A point at which two or more than two components are connected.

Kilowatt-hour(kwh): : A common unit of energy used mainly by utility companies.

Multimeter: : An instrument that measures voltage,current, and resistance.

Magnitude : The value of a quantity, such as the number of volts of voltage or the number of amperes of current.

Ohmmeter : An instrument for measuring resistance.

Ohm's law : A law stating that current is directly proportional to voltage and inversely proportional to resistance.

Parallel : The relationship in electronics circuits in which two or more current paths are connected between the same two points.

Passband : The range of frequencies passed by a filter.

Period : The time interval of one complete cycle of a periodic wave form.

Potentiometer : A three terminal variable resistor.

Primary winding : The input winding of a transformer, also called primary.

Rectifier : An electronic circuit that converts AC into pulsating DC, one part of a power supply.

Resistor : An electrical component designed specially to provide resistance. Limits the current value.

Rheostat : A two terminal variable resistor.

Root mean square(rms) : The value of a sinusoidal voltage that indicates its heating effect, also known as effective value. It is equal to 0.707 times the peak value.

Secondary winding : The output winding of a transformer, also called secondary.

Shell : The orbit in which an electron revolves.

Siemens : The unit of conductance.

Source : A device that produces electrical energy.

Step-down transformer : A transformer in which the secondary voltage is less than the primary voltage

Step - up transformer : A transformer in which the secondary voltage is greater than the primary voltage.

Superposition theorem : A method for the analysis of circuits with more than one source.

Switch : An electrical device for opening and closing a current path.

Thevenin's theorem : A method for simplifying a two terminal linear circuit to an equivalent with only a voltage source in series with a resistance or impedance.

Transformer : A device formed by two or more windings that are magnetically coupled to each other and provide a transfer of power electromagnetically from one winding to other.

Turn ratio : The ratio of turns in the secondary winding to turns in the primary winding.

Valence electron : An electron that is present in outer most shell of an atom.

Voltmeter : An instrument used to measure voltage.

Winding : The loops or turns of wire in an inductor.

PHY301 - Circuit Theory FAQs By www.virtualians.pk

Question: Define Shells and sub Shells

Answer:

Schematically energy levels of electrons in an atom are represented by shells and sub shells. Since in a sense electrons are moving around the nucleus, they possess orbital angular momentum. Now different orientations of angular momentum, having same magnitude, contribute to the same energy or energy level, we call that energy level as orbit or shell, and different orientations of orbital angular momentum which contribute to the same shell, as orbitals or sub shells. Shells are denoted as K, L, M, ..., and sub shells are denoted as s, p, d, ... and so on. Technically shells are represented by 'n' which can have values 1, 2, 3, ... (corresponds to K, L, M, ... respectively) and sub shells as 'l' having values 0, +1, +2, ..., (n-1), corresponding to s, p, d, ... respectively.

Beside orbital momentum, electrons also carry spin momentum due to their motion around their own axis. Electrons carry either $+1/2$ or $-1/2$ spins. Quantum theory also gives a simple formula for maximum occupancy of each shell, and it is, 'n' is the number of any shell. So K (n=1) can accommodate only 2 electrons, L 8, M 18, and so on.

This figure schematically shows shells and sub shells of an atom.

Check out this java applet to see different real images of shells and sub shells of hydrogen atom.

Question: What is Conductor?

Answer:

Conductors are material from which electric current can pass easily. Conductors have negligible resistance for the passage of current.

Every material in the world can be defined in terms of how well it conducts electricity. Certain things, such as cold glass, never conduct electricity. They're known as insulators. Materials which do conduct electricity, like copper, are called conductors. In the middle are materials known as semiconductors, which don't conduct as well as conductors, but can carry current. Last,

are materials called superconductors, which when brought down to very low temperatures turn into superhighways of current -- they conduct electricity without any resistance whatsoever.

All these different materials are made of atoms that look basically alike: a nucleus with electrons circling around them. What makes them so different when it comes to conducting electricity?

The difference comes down to nothing more than how the electrons are arranged around the nucleus. The laws of quantum physics say that there are only specific bands (or tracks) in which any electron can travel. There are some interesting facts about these bands. First of all, only a very specific number of electrons can travel in each one; once it's full, it's full. Second, which track an electron is in corresponds to how much energy that electron has. And third, some of the bands are closer to each other than others.

Different atoms have different numbers of electrons, and how those electrons are arranged in the bands defines whether a material made of those atoms will conduct.

In every atom, the electrons crowd down as close to the nucleus as possible, since the bands that are closest to the nucleus are also the ones that require the least energy. That means that the outermost track might not be completely filled. If it's not filled, then it's easy for an electron to jump from one atom into an empty space in the atom next door. Moving electrons, and therefore an electrical current. Atoms with empty spaces in the outermost electron bands are conductors.

Let's go to the next scenario, where the outermost track is completely filled. If the electrons in this track were given a little kick of energy -- say from a flash of light -- they might have enough energy to jump up to the next, empty track. But remember, some bands are close to each other, and some aren't. In atoms where the next track is close by, an energetic electron will have no problem jumping up a track. Suddenly, this electron is in a track with empty spaces, and electrons can move from atom to atom just as described above. Since these kinds of atoms can only conduct electricity sometimes -- when given this outside jolt of energy -- they're the semiconductors. Atoms with a full outside track which is very close to the next empty track are semiconductors.

If, however, the next potential track is too far away, then an electron can't jump to it even if it's given a jolt of energy. These electrons will always stick in their assigned track, never allowed to roam to another atom -- and never forming current. Atoms with a full outside track which is far from the next empty track are insulators.

Superconductors are a whole different breed, since no material known today super conducts except at very cold temperatures. Scientists are discovering materials that do super conduct

closer and closer to room temperature all the time, but no one is quite sure how that happens. However, John Bardeen, Leon Cooper, and Robert Schrieffer did come up with a theory for how the very coldest superconductors work, known as the BCS theory. In such materials, at low temperatures, the atoms vibrate in a way that forces the moving electrons closer together. Normally electrons don't like to huddle so close, since they're all electrically negative and therefore repel each other. But in superconductors, the electrons actually achieve almost an attraction for each other. The result is that as one electron moves, it pulls the next electron along right behind it. Electrons slip from atom to atom more easily than they ever do normally. Atoms which, at the right temperature, can make electrons attract instead of repel each other are superconductors.

Question: What is Polarity?

Answer: The negative polarity has been assigned to the static charge produced on rubber, amber and resinous materials in general. Positive polarity refers to the static charge produced on glass and other vitreous materials. On this basis the electrons in all atoms are basic particles of negative charge because their polarity is the same as the charge on rubber. Protons have positive charge because the polarity is the same as the charge on glass.

Question: What is Conventional Current? Why always electron flow from Negative to Positive ? Can it flow in reverse direction?

Answer:

The motion of positive charge, in the opposite direction from electron flow, is considered as conventional current.

Electricity was known of long before Benjamin Franklin. It was not understood very well, but it was known of. Scientists knew there were two kinds of electric charge. They knew there was electric current. Scientists believed that the opposite charges moved similarly in opposite directions. They defined one as positive and one as negative. They defined current to be in the direction of the positive charges. Later, they learned of their mistake.

Only the negative charges move freely in conductors. Electrons had been defined as the negative charges. Current had been defined "backwards". It was too late to redefine all of electrical physics, so the inconvenience holds to this day. The direction that the electrons move is opposite the direction that current points. Because of how electricity works, it isn't much of a problem.

Negative charge moving to the left through a wire has the same effect as positive charge moving to the right. So long as the total charge in the wire (protons and electrons) remains balanced, no trouble occurs.

See this interesting link

Question: What is the difference between series and parallel networks or circuits?

Answer:

Series circuits

A series circuit is a circuit in which resistors are arranged in a chain, so the current has only one path to take. The current is the same through each resistor. The total resistance of the circuit is found by simply adding up the resistance values of the individual resistors:

equivalent resistance of resistors in series : $R = R_1 + R_2 + R_3 + \dots$

A series circuit is shown in the diagram above. The current flows through each resistor in turn. If the values of the three resistors are:

$R_1 = 8\text{ohm}$, $R_2 = 8\text{ohm}$, and $R_3 = 4\text{ohm}$, the total resistance is $8+8+4=20\text{ohm}$

With a 10 V battery, by $V = I R$ the total current in the circuit is:

$I = V / R = 10 / 20 = 0.5 \text{ A}$. The current through each resistor would be 0.5 A.

Parallel circuits

A parallel circuit is a circuit in which the resistors are arranged with their heads connected together, and their tails connected together. The current in a parallel circuit breaks up, with some flowing along each parallel branch and re-combining when the branches meet again. The voltage across each resistor in parallel is the same.

The total resistance of a set of resistors in parallel is found by adding up the reciprocals of the resistance values, and then taking the reciprocal of the total:

equivalent resistance of resistors in parallel: $1 / R = 1 / R_1 + 1 / R_2 + 1 / R_3 + \dots$

A parallel circuit is shown in the diagram above. In this case the current supplied by the battery splits up, and the amount going through each resistor depends on the resistance. If the values of the three resistors are:

$R_1 = 8\text{ohm}$, $R_2 = 8\text{ohm}$, and $R_3 = 4\text{ohm}$, the total resistance is found by

$1/R = 1/8 + 1/8 + 1/4 = 1/2$. This gives $R = 2\text{ohm}$

With a 10 V battery, by $V = I R$ the total current in the circuit is: $I = V / R = 10 / 2 = 5 \text{ A}$.

The individual currents can also be found using $I = V / R$. The voltage across each resistor is 10 V, so:

$$I_1 = 10 / 8 = 1.25 \text{ A}$$

$$I_2 = 10 / 8 = 1.25 \text{ A}$$

$$I_3 = 10 / 4 = 2.5 \text{ A}$$

Note that the currents add together to 5A, the total current.

Circuits with series and parallel components

Many circuits have a combination of series and parallel resistors. Generally, the total resistance in a circuit like this is found by reducing the different series and parallel combinations step-by-step to end up with a single equivalent resistance for the circuit. This allows the current to be determined easily. The current flowing through each resistor can then be found by undoing the reduction process.

General rules for doing the reduction process include:

Two (or more) resistors with their heads directly connected together and their tails directly connected together are in parallel, and they can be reduced to one resistor using the equivalent resistance equation for resistors in parallel.

Two resistors connected together so that the tail of one is connected to the head of the next, with no other path for the current to take along the line connecting them, are in series and can be reduced to one equivalent resistor.

Finally, remember that for resistors in series, the current is the same for each resistor, and for resistors in parallel, the voltage is the same for each one.

Question: What is kirchhoff's current Law (KCL) and Voltage Law (KVL)?

Answer:

Kirchhoff's Current Law

This fundamental law results from the conservation of charge. It applies to a junction or node in a circuit -- a point in the circuit where charge has several possible paths to travel.

In Figure 1, we see that I_A is the only current flowing into the node. However, there are three paths for current to leave the node, and these current are represented by I_B , I_C , and I_D .

Once charge has entered into the node, it has no place to go except to leave (this is known as conservation of charge). The total charge flowing into a node must be the same as the the total charge flowing out of the node. So,

$$I_B + I_C + I_D = I_A$$

Bringing everything to the left side of the above equation, we get

$$(I_B + I_C + I_D) - I_A = 0$$

Then, the sum of all the currents is zero. This can be generalized as follows

$$\sum I = 0$$

Note the convention we have chosen in Fig 1: current flowing into the node are taken to be negative, and currents flowing out of the node are positive. It should not really matter which you

choose to be the positive or negative current, as long as you stay consistent. However, it may be a good idea to find out the convention used in your class.

Kirchhoff's Voltage Law

Kirchhoff's Voltage Law (or Kirchhoff's Loop Rule) is a result of the electrostatic field being conservative. It states that the total voltage around a closed loop must be zero. If this were not the case, then when we travel around a closed loop, the voltages would be indefinite. So

$$\sum V = 0$$

In Fig 2 the total voltage around loop 1 should sum to zero, as does the total voltage in loop2. Furthermore, the loop which consists of the outer part of the circuit (the path ABCD) should also sum to zero.

Question: What is meant by "Loop" ?

Answer: A loop is a closed path formed by starting at a node, passing through a set of nodes, and returning to the starting node without passing through any node more than once.

Question: What is meant by "Junction" ?

Answer: junction and node both are the same things, node is also called junction.

It is the contact point between two or more than two elements or we can say that it is the point of connection between circuit.

See this figure. This figure will help you to understand

Question: What is reference node?

Answer: The reference node is commonly called the ground since it is assumed to have zero potential. The choice of a reference node is completely arbitrary, but the node with the largest number of components or voltage source connected to it is usually most convenient. Generally when we are choosing reference node on paper we will consider the bottom node of the circuit as reference node. See this figure

Question: What is Thevenin's theorem ?

Answer:

Thevenin's Theorem

The Thevenin's theorem states that any linear network can be represented by a voltage source in series with an impedance. We will generally use a simpler version: any circuit made up of voltage sources, current sources and resistors can be replaced by a voltage and a resistor.

When applying the Thevenin Theorem there are three cases.

Case 1: Only independent sources . In the typical case, there are no dependent sources in the circuit to be Thevenized. To find the Thevenin equivalent, first find the open circuit voltage, V_{oc} , this is the Thevenin voltage. To find the Thevenin resistance, set all sources to zero and find the resistance of the resulting circuit

Consider again the circuit from above,

and try to find the Thevenin circuit at the terminals (i.e., across the 1k resistor). From the discussion of superposition, we know the open circuit voltage, V_{oc} , is 1.666 volts. The Thevenin resistance, R_T , is found by finding the equivalent resistance of the circuit with all source set to zero, as shown below

Obviously the Thevenin resistance, R_T , is $1k \parallel 500 = 333\Omega$. Therefore the resulting circuit is:

Case 2: Independent and Dependent Sources. If the circuit to be Thevenized has both dependent and independent source, the method described above cannot be used to find the Thevenin resistance. Instead, you must find the short circuit current, I_{sc} (current through short circuit at terminals). Then the Thevenin resistance is given by $R_T = V_{oc}/I_{sc}$.

Question: What are the main concepts of Superposition Theorem?

Answer:

The superposition theorem is a method of solving circuits, often used in circuits with more than one emf source. In a network containing multiple independent source, each source can be applied independently with the remaining source turned off. In order to use one source at a time, all other sources are "killed" temporarily.

This means disabling the source so that it can not generate voltage or current, without changing the resistance of the circuit.

Keep these point in mind .

- (i) A voltage source such as a battery is killed or turned off by assuming a short circuit across its potential difference.
- (ii) A current source is killed or turned off by replacing it with an open circuit.
- (iii) The result obtained by applying each source independently are then added together algebraically to obtain a solution.

Question: What is a short circuit?

Answer: A circuit element with resistance approaching zero. ($R = 0$)

See this figure.

In short circuit voltage source has a closed path across its terminals. In other words connecting both the terminals of voltage source without the resistance between them. The result is too much current in the circuit. For instance, a short across the conducting wires for a bulb produces too much current in the wires but no current through the bulb. Then the bulb is shorted out. The bulb is not damaged, but the wires can become hot enough to burn.

Question: What is ground?

Answer: Ground

Every electrical circuit has a point of reference to which all circuit voltages are compared. This reference point is called ground, and circuit voltages are either positive or negative with respect to ground. Connections to ground that are made for safety reasons refer to earth ground. When voltage measurements are taken, the difference of potential between a point in the circuit and a ground point is measured by the voltmeter. This type of ground is referred to as chassis or common ground.

Earth Ground.

Initially, ground referred to the earth itself and since has represented a point of zero potential or zero volts. A short circuit within a device that connects live voltage to the frame could cause a serious shock to anyone touching it. However, if the frame is connected to earth ground, it is held at the safe potential of zero volts, as the earth itself absorbs the voltage.

Question: What is the difference b/w independent and dependent sources?

Answer: Independent Voltage Source

An independent voltage source is a two terminal element that maintain a specified voltage between its terminals regardless of the current through it. The general symbol for an independent source is circle. Dependent Voltage Source

Dependent or controlled voltage source generate a voltage that is determined by a voltage at a specified location in the circuit. Such sources are very important because they are integral part of our mathematical models. The general symbol for dependent voltage source is diamond

Question: What is meant by "Super Node" ?

Answer: A node which emerges as a result of combination of two ordinary nodes around a voltage source is called a super node.

Question: How can we write Constraint or Coupling equation for super node?

Answer: Constraint or coupling equation describes a super node mathematically, instead of writing equations individual ordinary nodes of the super node. The difference in potential between the two nodes is equal to voltage source between two nodes. Now to write constraint equation first write the node which is towards +ve terminal of voltage source then subtract the node which is towards -ve side of voltage source then equal this difference to value of voltage source.

Question: What is Norton's Theorem ?

Answer: • Norton's Theorem is a way to reduce a network to an equivalent circuit composed of a single current source, parallel resistance, and parallel load.

- Steps to follow for Norton's Theorem:
- (1) Find the Norton source current by removing the load resistor from the original circuit and calculating current through a short (wire) jumping across the open connection points where the load resistor used to be.
- (2) Find the Norton resistance by removing all power sources in the original circuit (voltage sources shorted and current sources open) and calculating total resistance between the open connection points.
- (3) Draw the Norton equivalent circuit, with the Norton current source in parallel with the Norton resistance. The load resistor re-attaches between the two open points of the equivalent circuit.
- (4) Analyze voltage and current for the load resistor following the rules for parallel circuits.

Question: What is peak voltage?

Answer: Peak voltage tell you how far the voltage swings, either positive or negative, from the point of reference. Peak voltage is only a moderately useful way of measuring voltage when trying to express the amount of work that will be done when driving a specified load. Some

manufacturers use peak voltage to get the power output ratings of their amplifiers. For Figure Click Here

Question: What is RMS voltage ?

Answer: RMS voltage is absolutely the most common way to measure/quantify AC voltage. It is also the most useful. Because AC voltage is constantly changing and is at or near the highest and lowest points in the cycle for only a tiny fraction of the cycle, the peak voltage is not a good way to determine how much work can be done by an AC power source (e.g. your amplifier, a wall outlet in your house...). DC voltage is constant. Its voltage level can be plugged directly into the formulas for power and you will get an accurate image of its ability to do work. RMS voltage will give you the same ability to predict how much work will be done by an AC voltage. The RMS voltage of a Pure sine wave is approximately $.707 \times \text{peak voltage}$. If you read voltage with a voltmeter you are generally given the RMS voltage of the wave form. Some meters display an 'average' voltage which is very close to RMS. When reading voltage with a voltmeter, the display indicates the RMS or average voltage not the peak or peak-peak voltage.

*If the waveform isn't a pure sine wave (like a square wave or a signal with mixed sine waves of different frequencies or music), multiplying the peak times $.707$ will not give an accurate RMS value and therefore will not give an accurate indication of the work that the waveform can produce when driving a load. For more complex signals, you need a meter that will calculate the RMS value from a set of samples taken at regular intervals. For Figure Click Here

Question: What is an Ideal diode?

Answer: The diode can be considered to be a one way street, that is it conducts electricity well in one direction but hardly any in the opposite direction. An ideal diode has no resistance in the forward direction and infinite resistance in the reverse direction. An ideal diode is like a light switch in your home. When the switch is closed, the circuit is completed; and the light turns on. When the switch is open, there is no current and the light is off.

However, the diode has an additional property; it is unidirectional, i.e. current flows in only one direction (anode to cathode internally). When a forward voltage is applied, the diode conducts; and when a reverse voltage is applied, there is no conduction. A mechanical analogy is a rat trap, which allows motion in one direction only.

Question: Define Atomic Structure.

Answer:

Atoms consist of negative charges (electrons) revolving around the nucleus, which itself consists of positive charges (protons) and neutral particles (neutrons). Number of electrons and a proton in an isolated atom remain equal and are called Atomic Number of that atom. Hence the equal negative charge shields the positive charge at the center and atom as a whole become neutral.

According to the quantum theory of atoms, energies, referred as energy levels, of electrons in an isolated atom are not continuous but discrete. This means that electrons require certain minimum amount of energy to increase their energy level, or in other words they can't increase their energy unless they are given that minimum energy. Check out this very interesting discussion regarding energy levels.

Question: What is Luminous intensity?

Answer:

Luminous intensity is an expression of the amount of light power emanating from a point source within a solid angle of one steradian. For reference, a frequency of 540 terahertz (540 THz or 5.40×10^{14} Hz) is specified. The quantities used to express luminous intensity are arcane to most non-scientists. A frequency of 540 THz corresponds to a wavelength of about 555 nanometers (nm), which is in the middle of the visible-light spectrum, and is generally accepted as the frequency and wavelength at which the average human eye is most sensitive. A steradian is the standard unit solid angle; a sphere encloses 4π (approximately 12.57) steradians.

Decades ago, luminous intensity was measured in terms of a unit called the candle. This expression arose from the fact that one candle represented approximately the amount of visible radiation emitted by a candle flame. This was an inexact specification, because burning candles vary in brilliance. So, for a time, a specified amount of radiation from elemental platinum at its freezing temperature was used as the standard. Late in the 20th century, the candela was defined and adopted as the standard unit of luminous intensity. One candela (1 cd) is the magnitude of an electromagnetic field (EM-field), in a specified direction, that has a power level equivalent to a visible-light field of $1/683$ watt (1.46×10^{-3} W) per steradian at 540 THz.

Question: What are Insulators?

Answer: Material in which electrical charges do not move freely from place to place. (wood, paper, plastic, glass, rubber).

Question: What is Centripetal force and centrifugal force?

Answer: Centripetal force and centrifugal force, action-reaction force pair associated with circular motion. According to Newton's first law of motion, a moving body travels along a straight path with constant speed (i.e., has constant velocity) unless it is acted on by an outside force. For circular motion to occur there must be constant force acting on a body, pushing it toward the center of the circular path. This force is the centripetal (center-seeking) force. For a planet orbiting the sun, the force is gravitational; for an object twirled on a string, the force is mechanical; for an electron orbiting an atom, it is electrical. The magnitude F of the centripetal force is equal to the mass m of the body times its velocity squared v^2 divided by the radius r of its path: $F = mv^2/r$. According to Newton's third law of motion, for every action there is an equal and opposite reaction. The centripetal force, the action, is balanced by a reaction force, the centrifugal (center-fleeing) force. The two forces are equal in magnitude and opposite in direction. The centrifugal force does not act on the body in motion; the only force acting on the body in motion is the centripetal force. The centrifugal force acts on the source of the centripetal force to displace it radially from the center of the path. Thus, in twirling a mass on a string, the centripetal force transmitted by the string pulls in on the mass to keep it in its circular path, while the centrifugal force transmitted by the string pulls outward on its point of attachment at the center of the path. The centrifugal force is often mistakenly thought to cause a body to fly out of its circular path when it is released; rather, it is the removal of the centripetal force that allows the body to travel in a straight line as required by Newton's first law. If there were in fact a force acting to force the body out of its circular path, its path when released would not be the straight tangential course that is always observed.

Question: What is the difference between open and closed circuit?

Answer:

OPEN CIRCUIT:

When any part of the path is broken, the circuit is open because there is no continuity in the conducting path. The resistance of an open circuit is infinitely high. The result is no current in open circuit.

CLOSED CIRCUIT

is a circuit which provides complete or continuous path for current flow., in other words current flows from one side of the applied voltage source, through the external circuit, and

returns to the other side of voltage source. in closed circuit there must be a resistance in the path of current.

Question: What is the difference between resistance and conductance?

Answer:

RESISTANCE is the opposition to current flow. Its unit is ohm. Resistors are perhaps the most common component in electronic circuits. Their main function is to reduce the current I to the desired value or to provide the desired voltage in a circuit

CONDUCTANCE is the ability to conduct current. It is the reciprocal of resistance $= I/R$. The unit is the Siemens's).

Question: Define Potential difference and Current.

Answer:

Current (measured in Amperes) is actually the flow of electrons (charge). Its like finding the quantity of water flowing through a pipe of certain area in unit time. So imagine the cross-section of a wire having electrons passing through. Now if you count the no. of electrons passing through the cross-section in one second that would be exactly the quantity what is called current?

A volt is the measure of Potential Difference across circuit elements (battery, resistor etc.). Consider two cylinders of water one having less water than other, now link the two cylinders with a pipe, so that the water from one cylinder can pass to other. Now what will be the direction of water?..... YES. you are right. water will flow from the cylinder having more water to one that has less water. this will result in decreasing of water level from where it is flowing and increase in water level in cylinder to which water is flowing. When both levels come to same height, water will stop to flow. Technically this happens due to the difference in the potential energies of water in two cylinders. Water will flow only when there is a potential difference, and when it is zero (i.e. when two levels become equal) water will stop to flow. Same thing happens when we create a potential difference across a wire, on one side of wire there are more electrons, creating a negative charge region, while on the other end electronic concentration is low, becoming a positive region as compared to the negative region. This potential difference will make electrons to flow from negative to the positive side. thus resulting in non zero current. In case of zero potential difference (i.e. when both side have equal no. of electrons making the two sides at the same potential), no electron will feel a 'push' to either end, thus results in zero current.

This is the basic relation between currents and volts, with currents acting as 'effect' and volts as 'cause'. Now since under applied voltages, electrons flow through wire, that means by applying voltages we give electric energy (measured in Joules) to the wire,

Check this interesting link to get a deep insight into charge flow.

Question: What is passive sign Convention?

Answer: The passive sign convention: When we observe that positive current enters the positive terminal of a component, we say that the component obeys the passive sign convention (PSC). Therefore, when the passive sign convention is being obeyed, it indicates that a component is dissipating energy (or power) as charge is being displaced from a higher potential to a lower potential. One way to think about this is using another water analogy. If we splice a garden hose in a length of fire hose we create a pressure potential. The high pressure side is the one where the water is entering. The garden hose is analogous to a resistor which impedes the flow of current. Thus, when current flows through a resistor, a higher voltage potential will exist on the incoming current side. In this example, the garden hose and the resistor are obeying the passive sign convention. To reiterate, the PSC is obeyed when the current enters an element's positive terminal and exits at the negative terminal. The passive sign convention is usually used for assigning reference marks for voltage drops across, and currents through, resistors, but we frequently assign a current and associated direction for a current through a voltage source in violation of the PSC. By the same token, we frequently define a voltage and its associated reference marks across a current source in violation of the PSC. To successfully apply Ohms law, you must consistently observe the proper relationship between applied voltage and the direction of current flow. See this image

Question: What is resistance?

Answer:

Resistance

Resistance is a term that describes the forces that oppose the flow of electron current in a conductor. All materials naturally contain some resistance to the flow of electron current. We have not found a way to make conductors that do not have some resistance.

If we use our water analogy to help picture resistance, think of a hose that is partially plugged with sand. The sand will slow the flow of water in the hose. We can say that the plugged hose has more resistance to water flow than does an unplugged hose. If we want to get more water out

of the hose, we would need to turn up the water pressure at the hydrant. The same is true with electricity. Materials with low resistance let electricity flow easily. Materials with higher resistance require more voltage (EMF) to make the electricity flow.

The scientific definition of one ohm is the amount of electrical resistance that exists in an electrical circuit when one amp of current is flowing with one volt being applied to the circuit.

Resistance depends upon the following factors

- a) length
- b) area
- c) temperature

$$R = r * L / A$$

Is resistance good or bad?

Resistance can be both good and bad. If we are trying to transmit electricity from one place to another through a conductor, resistance is undesirable in the conductor. Resistance causes some of the electrical energy to turn into heat so some electrical energy is lost along the way. However, it is resistance that allows us to use electricity for heat and light. The heat that is generated from electric heaters or the light that we get from light bulbs is due to resistance. In a light bulb, the electricity flowing through the filament, or the tiny wires inside the bulb, cause them to glow white hot. If all the oxygen were not removed from inside the bulb, the wires would burn up.

An important point to mention here is that the resistance is higher in smaller wires. Therefore, if the voltage or EMF is high, too much current will follow through small wires and make them hot. In some cases hot enough to cause a fire or even explode. Therefore, it is sometimes useful to add components called resistors into an electrical circuit to slow the flow of electricity and protect of the components in the circuit.

Resistance is also good because it gives us a way to shield ourselves from the harmful energy of electricity.

Question: What is inductance?

Answer: INDUCTANCE:

The characteristic of an electrical circuit that opposes a change in current. The reaction (opposition) is caused by the creation or destruction of a magnetic field. When current starts to flow, magnetic lines of force are created. These lines of force cut the conductor inducing a counter emf in a direction that opposes current.

Question: What is Power dissipation?

Answer:

Power dissipation

When a current flows through a component, that component will heat up. This process is called power dissipation and is measured in Watts. The power dissipation of a device can be calculated very easily:

$$P = V \cdot I$$

Let's calculate the power dissipation of a 100 ohms resistor connected to a 9V battery.

The voltage across the resistor will be 9V. The current is $9V/100\text{ohms} = 90\text{mA}$. So the power dissipation will be: $9V \cdot 90\text{mA} = 810\text{mW}$.

It is very important to calculate the power dissipation of the components in your design. A regular resistor has a maximum dissipation rating of 0.25W (= 250mW). If you would have used such a resistor in the example above, it would have blown. A 1W resistor is a good choice.

Since it's so important, let's create an equation with which we can easily calculate the power dissipation of a resistor. We know:

$$P = V \cdot I \text{ ----(1)}$$

$$V = I \cdot R \text{ ----(2)}$$

$$I = V / R \text{ ----(3)}$$

Substituting (2) in (1) and (3) in (1) respectively results in:

$$P = I^2 \cdot R$$

$$P = V^2 / R$$

With these equations you can easily calculate the power dissipation when you connect a DC voltage source to a resistor.

Question: What is an open circuit?

Answer: An open circuit: A circuit element with resistance approaching infinity.

($R \rightarrow \infty$).

See this figure.

.

Question: What is meant by " Branch " ?

Answer: Branches A branch is any path in the circuit that has a node at each end and contains at least one voltage source or resistor but contains no other nodes. This circuit contains 6 branches, denoted B1, ..., B6. If branch B4 did not contain a resistor then it could be deleted and nodes N2 and N3 could be considered one and the same node. See this figure.

Question: What are the rules for parallel and series resistances in a network or circuit?

Answer: Keep these two things in mind when you are solving such circuits.

Resistors are in series if they carry exactly the same current (share one common node).

Resistors are in parallel if they have the same voltage across them and are connected exactly between the same two nodes.

Question: What is source transformation?

Answer:

Source Transformation method is used to find unknown value of a voltage or current in a circuit.

If we have any source embedded within a network, say this source is a current source having a value I & there exists a resistance having a value R , in parallel to it. We can replace it with a voltage source of value $V=IR$ in series with same resistance R .

The reverse is also true that is a voltage source V , in series with a resistance R can be replaced by a current source having a value $I=V/R$

In parallel to the resistance R .

Parameters within circuit, for example an output voltage remain unchanged under these transformations.

Question: What is a function of diode?

Answer: Diodes allow electricity to flow in only one direction. The arrow of the circuit symbol shows the direction in which the current can flow. Diodes are the electrical version of a valve and early diodes were actually called valves. To see the figure please click [here](#). A diode is made of two different types of semiconductors right next to each other. One side is easy for electrons to travel through; one side is much tougher. It's something like trying to swim through a pool filled with water and then a pool filled with mud: swimming through water is easy; swimming through mud is next to impossible. To an electron some semiconductors seem like water, some like mud.

One side of the semiconductor boundary is like mud, one like water. If you try to get electricity to move from the mud side to the water side, there's no problem. The electrons just jump across the boundary, forming a current. But try to make electricity go the other way and nothing will happen. Electrons that didn't have to work hard to travel around the water side just don't have enough energy to make it into the mud side. (In real life, there are always a few electrons that can trickle in the wrong direction, but not enough to make a big difference.

There are a number of different electronic devices which tend to be called diodes. Although they're made differently they all have three things in common.

1. They have two leads like a resistor.

2. The current they pass depends upon the voltage between the leads.

3. They do not obey Ohm's law!

As an example we will use a typical diode called a pn-junction. This allows us to explain behavior of diodes. Remember, however, that there are other sorts of diodes which are built differently but show the same general behavior.

We create a pn-junction by joining together two pieces of semiconductor, one doped n-type, the other p-type. This causes a depletion zone to form around the junction (the join) between the two materials. This zone controls the behavior of the diode.

Question: What is meant by Forward Voltage Drop ?

Answer: Electricity uses up a little energy pushing its way through the diode, rather like a person pushing through a door with a spring. This means that there is a small voltage across a conducting diode, it is called the forward voltage drop and is about 0.7V for all normal diodes which are made from silicon.

Question: What is meant by "Reverse Voltage" in a diode?

Answer: Reverse Voltage When a reverse voltage is applied a perfect diode does not conduct, but all real diodes leak a very tiny current of a few μA or less. This can be ignored in most circuits because it will be very much smaller than the current flowing in the forward direction. However, all diodes have a maximum reverse voltage (usually 50V or more) and if this is exceeded the diode will fail and pass a large current in the reverse direction, this is called breakdown.

Question: What is horse power and why is it called horse power?

Answer: The term horsepower was invented by the engineer James Watt. Watt lived from 1736 to 1819 and is most famous for his work on improving the performance of steam engines. We are also reminded of him every day when we talk about 60-watt light bulbs. What horsepower means is this: In Watt's judgment, one horse can do 33,000 foot-pounds of work every minute.

checkout this very interesting link regarding story behind horsepower.